

MSc in Computing - Team Project

User Evaluation Report

ClauseGuard(Group 3)

Team Members
A00047237 – Siddharth Varma Raja
A00047190 – Prasad Nitin Pansare
A00047193 – Shubham Tatyasaheb Patil
A00046404 – Kameshwara Karthik Ayyalasomayajula
A00046791 – Jude Fernandes

1. Proposed Hypothesis (What Questions Are You Asking?)

ClauseGuard is a web application that checks Irish residential leases against Residential Tenancies Board (RTB) rules. It rates each clause by risk, explains it in plain English (Figure 1), answers questions through a chatbot, drafts an email to the landlord and produces a PDF report. Its users are renters, often students or new to Ireland, with no legal training.

Our evaluation asked three questions:

- **RQ1 Usability:** can first-time users complete the core tasks (check a lease, judge its overall risk, find a specific problem, understand the risk colours, email the landlord, save a report) without help?
- **RQ2 Understanding:** do the plain-English explanations let people without legal knowledge see why a clause is a problem and reach the right decision about signing?
- **RQ3 Trust and adoption:** do users trust the analysis enough to act on it, and would they use it before signing a real lease?

Our hypothesis was that renters, including those who have never signed an Irish lease, can use ClauseGuard to find the problem clauses in a lease and decide correctly whether to sign, and would trust and reuse it. These questions matter because a misunderstood or untrusted result could lead a renter to sign an unfair lease.

Automated detection of unfair clauses is usually evaluated by precision and recall against expert labels. CLAUDETTE (Lippi et al., 2019), which detects potentially unfair clauses in online terms of service, is a well-known example. Such evaluations measure accuracy, not whether lay users understand or act on the output. Usability is typically evaluated with inspection methods, such as heuristic evaluation (Nielsen and Molich, 1990; Nielsen, 1994) and the cognitive walkthrough (Wharton et al., 1994), together with task-based testing with small groups, since about five users find most usability problems (Nielsen and Landauer, 1993). Inspection methods depend on evaluator judgement and can miss or over-predict problems, and small tests cannot support statistical claims, so we combined four methods.

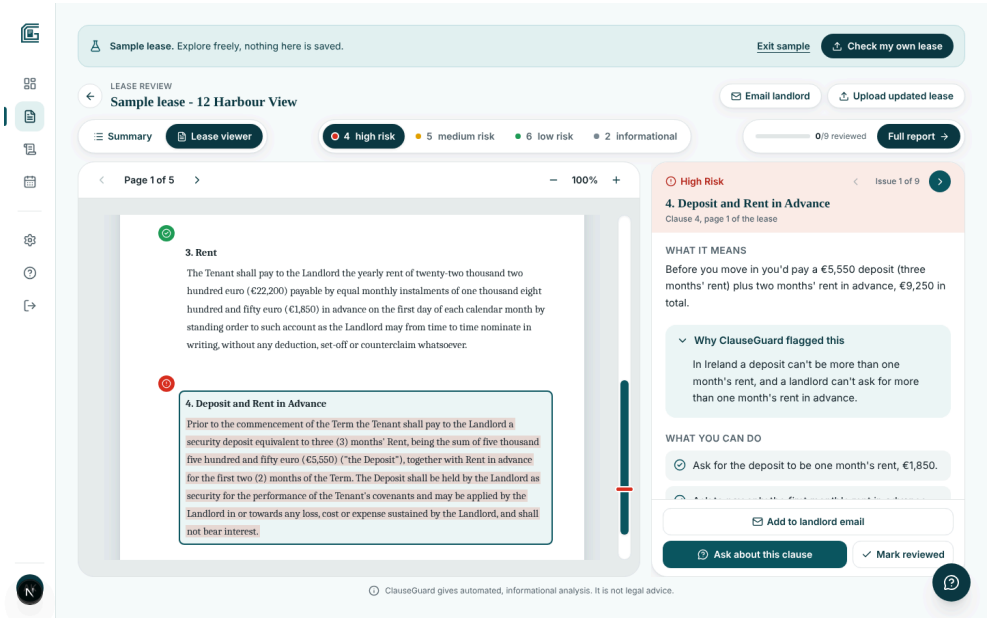


Figure 1. ClauseGuard Lease viewer: a High Risk clause in the sample lease, with its plain-English explanation.

2. Experimental Method

2.1 Overview

We used a mixed-methods design with four methods, run in sequence, so that the team's predictions could be checked against real users:

- **Cognitive walkthrough:** the team stepped through 6 core tasks as a first-time renter persona (Priya, 22, an international student) and answered the four walkthrough questions at every step.
- **Expert review:** an external reviewer evaluated the site against Nielsen's 10 heuristics and rated each problem's severity from 0 to 4.
- **Usability test:** 4 participants completed 7 goal-based tasks (sign up plus the 6 core tasks) within a realistic scenario, thinking aloud while we observed.
- **Survey:** 25 participants completed task-based sections in a Google Form while using the live site.

Everyone used the built-in sample lease, so answers could be checked against a known key: ClauseGuard rates the sample lease "Not yet" to sign, with four High Risk clauses, including a deposit of three months' rent. There was no "without ClauseGuard" control condition. Instead, we compared results with the answer key, with the walkthrough's predictions, and between participants who had and had not signed an Irish lease. The metrics were task success, time on task, ease (1–7), comprehension (answer card out of 3), survey ratings (1–5), willingness to use, and problems found by severity.

2.2 Data Collection

We collected both objective and subjective data. Objective data were task success (✓ or ✗), time on task, answer-card scores and the problems found by each method with their severity. Subjective data were ease ratings after each task (the Single Ease Question; Sauro and Dumas, 2009), the survey's 1–5 ratings and multiple-choice answers, think-aloud comments and free-text answers. Each measure maps to a research question: task success and time to RQ1; explanation clarity (Q6), understanding why a clause was flagged (Q7) and the answer card to RQ2; and trust (Q13) and willingness to use (Q14) to RQ3.

2.3 Selected Subjects

The survey had 25 participants, recruited through classmates, friends and renter contacts (a convenience sample). Twelve were aged 24 or under, 7 were 25–29, 3 were 30–39 and 3 were 40 or over. Twelve had signed an Irish lease before and 13 had not, which matches ClauseGuard's target users: students and first-time renters. Four further people, none of them on the team, took part in the usability test, using a laptop (2), a tablet (1) and a phone (1). Three external reviewers carried out the expert review. The sample is small, young and self-selected, so it represents student renters better than renters in general.

2.4 Data Analysis

We counted successes per task with median time and mean ease, averaged survey ratings and compared the two lease-experience groups, and grouped free-text answers into themes (Braun and Clarke, 2006). With 4 test participants we report counts, not significance tests.

Table 1. Usability test results (4 participants; times are medians of all attempts).

Task (usability test)	Success	Median time	Mean ease (1–7)
1. Sign up / log in	3 of 4	0:50	6.0
2. Check the lease	4 of 4	1:18	5.5
3. Is it safe to sign?	4 of 4	3:10	5.0
4. Deposit problem	2 of 4	4:42	4.3
5. What the colours mean	4 of 4	0:58	6.3
6. Message the landlord	1 of 4	2:15	2.8
7. Save a copy (PDF)	3 of 4	2:31	5.5

RQ1: participants completed 21 of 28 task attempts (75%). Drafting the landlord email was the weakest task: in 3 of 4 sessions the draft kept loading and no error appeared. The deposit task failed twice because the clause wording was hard to follow. The walkthrough's main prediction, that users would not understand the risk colours, was not confirmed: all 4 learned them from the summary labels. RQ2: all 4 correctly decided not to sign the lease as it is.

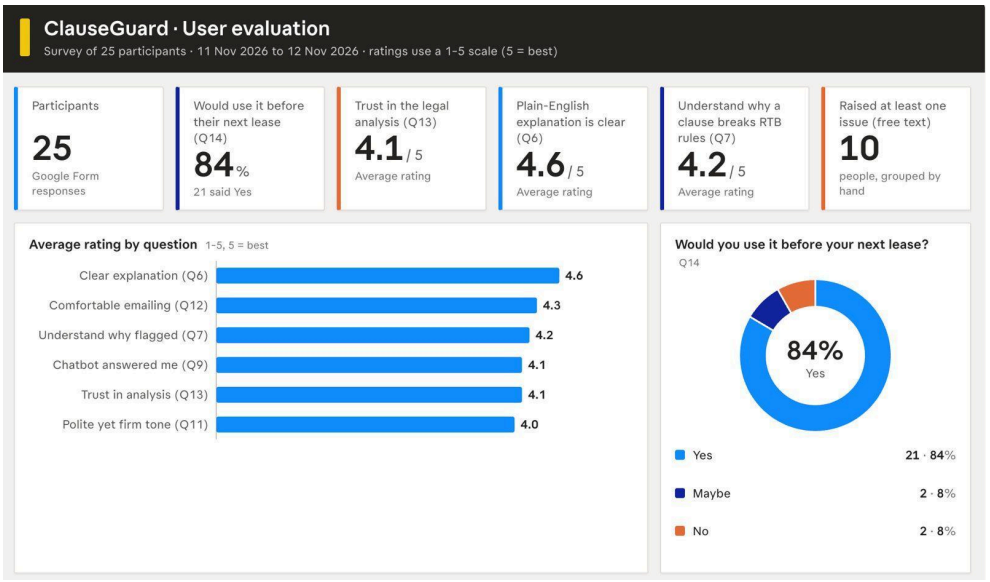


Figure 2. Survey results (25 participants, 1–5 scale).

In the survey, explanations were rated clear (Q6: 4.6/5) and participants understood why clauses were flagged (Q7: 4.2). Trust was 4.1 (Q13), and 21 of 25 (84%) would use ClauseGuard before their next lease (Q14). Progress during upload was very clear to 76% (Q4), but only 64% found the wait acceptable (Q5). Participants who had never signed an Irish lease rated every question lower, most of all understanding why a clause was flagged (3.6 vs 4.8) and comfort sending the email (3.8 vs 4.8; Figure 3): the renters who need the explanations most understood them least.

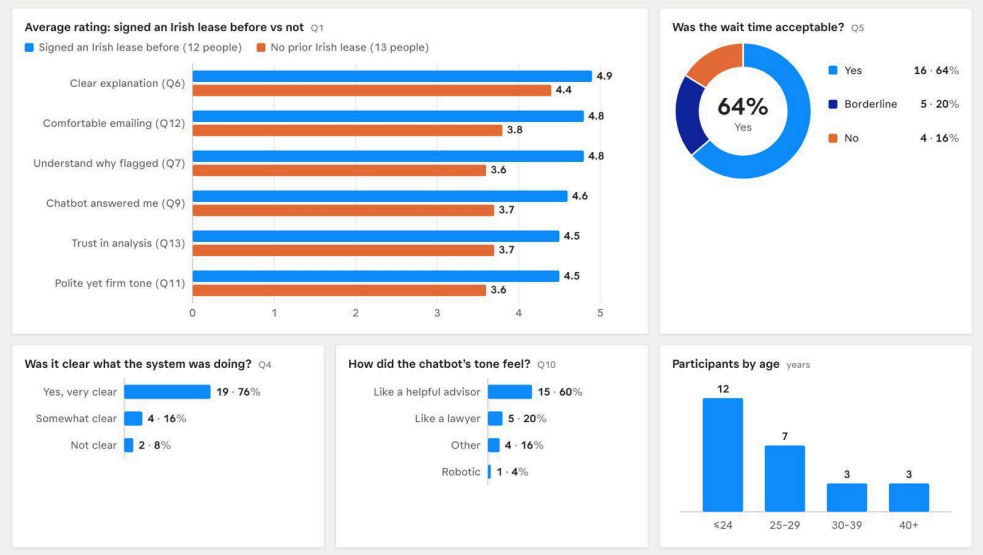


Figure 3. Survey ratings by lease experience, wait time (Q5), clarity of progress (Q4), chatbot tone (Q10) and age.

The expert found 16 problems: 9 consistency problems (H4) and 4 design problems (H8). Two were major: there is no way to cancel or leave the upload pop-up, and the yellow deposit card wrongly suggests a warning. The walkthrough found 9 problems, two of them major. Table 2 compares the issues found by more than one method.

Table 2. Issues found by more than one method.

Issue	Walkthrough	Expert review	Usability test	Survey
Landlord email slow or failing	Not testable locally	–	3 of 4 failed	3 mention email not working; 1 slow loading
Waiting during upload	Progress on a timer (minor)	No cancel (major)	–	36% wait not fully acceptable
Deposit clause and card	Yellow vs red (cosmetic)	Yellow implies warning (major)	2 of 4 failed deposit task	1 legal wording confusing
Too much information	–	4 design (H8) problems	2 of 4 overwhelmed by general clauses	3 want simpler navigation
Trust	–	Few trust-building sections	1 hesitated on the landing page	3 want official references
Risk colours and icons	Hover not obvious (major)	Count not coloured (cosmetic)	Not seen: 4 of 4 succeeded	–

2.5 Practical Setup

The survey ran online from 08 to 12 November 2026. Participants opened the live site (www.clauseguard.ie) on their own device and followed the form's six sections: About You, then four task sections (upload the sample lease, review the flagged clauses, ask the chatbot "Can my landlord actually enforce this?", generate a landlord email), then Overall Feedback. Each section showed the task on screen, followed by its questions. It took about 15 minutes, and the responses were anonymised before analysis.

The usability tests took place in person, on the participant's own laptop, tablet or phone, in a quiet room. We read out a scenario card ("You've found a flat in Dublin…

decide whether you should sign it"), then gave the tasks one at a time as goals, not buttons. Participants were asked to think aloud, and we gave no help; a task was marked failed if they gave up. One team member observed and recorded the result, time, ease and comments on a note sheet. At the end, participants filled in a paper answer card.

The expert received a written brief with the steps, the 10 heuristics, the severity scale and a findings table, and returned it in writing. The walkthrough was done by the team on a local build with the sample lease.

3. Conclusions

The hypothesis was largely supported. Participants completed 75% of the tasks unaided, all four test participants made the right signing decision, explanations were rated clear (4.6/5), and 84% would use ClauseGuard before their next lease. However, the landlord email failed in most sessions, waiting during upload frustrated a third of the survey participants, and people new to Irish leases understood the explanations noticeably less well (3.6 vs 4.8).This study measured users, not model accuracy. The clause classifier was tested separately: the hybrid model scores 0.63 macro-F1 on 389 clauses from five leases, against 0.58 for the keyword baseline.

The priorities are to:

- make the email reliable, with a fallback draft and a clear error message;
- let the upload run in the background, or be cancelled;
- show deposit facts in a neutral colour;
- reduce the general clauses shown;
- link explanations to official RTB guidance to build trust.

Using all four methods was valuable: the walkthrough over-predicted the colour problem and missed the email failure, which only real users revealed. A future study should:

- compare reading the lease alone with using ClauseGuard;
- test 5–8 participants, including more phone users;
- use unseen leases;
- add the standard SUS questionnaire;
- include a housing-law expert such as Threshold;
- pilot the survey, so every question is analysed and no email addresses are collected.

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Appendix A: Cognitive Walkthrough

The team walked through 6 core tasks as the persona Priya (22, an international student renting in Ireland for the first time, no legal knowledge, using a laptop). At every step we asked: Will she try to do this? Will she see the control? Will she connect it to her goal? Will she know it worked? Tasks 1, 3 and 5 passed with no major problems; tasks 2, 4 and 6 each had a step where a first-time user could get stuck.

Table A1. Walkthrough outcome per task.

Task	Outcome
1. Get ClauseGuard to check a lease	Passed. Progress steps run on a timer, so a slow server can look stuck (step 1.3).
2. Find out whether the lease has serious problems	Partly. The summary counts the clauses needing attention but does not say plainly whether to sign; the verdict is only on the Full report (step 2.1).
3. Find the deposit problem and why it matters	Passed. The deposit warning is yellow although the clause is High Risk (step 3.1).
4. Find out what a red circle means	Partly. Nothing shows the icon can be hovered, and hover does not work on phones (step 4.1).
5. Prepare a message to the landlord	Passed. The email app opens with no recipient (step 5.4).
6. Save a copy of the results	Partly. The download sits behind “Full report”, and Download PDF gives no feedback or error (steps 6.1–6.2).

Table A2. Problems found by the walkthrough (1 cosmetic, 2 minor, 3 major, 4 blocks the task).

#	Problem	Step	Severity (1–4)
1	Risk icons give no hint they can be hovered, and do not work on phones	4.1	3
2	Download PDF shows no feedback while the PDF is made, and no message if it fails	6.2	3
3	The PDF says “Prepared for: User” instead of the person’s name	6.2	2
4	The only way to download is through “Full report”, which does not mention a download	6.1	2
5	The summary does not plainly say whether it is safe to sign	2.1	2
6	Upload progress runs on a timer, so it can look stuck	1.3	2

7	The deposit warning is yellow, but the clause is High Risk (red)	3.1	1
8	The email app opens with no recipient	5.4	1
9	Page tours start by themselves and cover part of the screen on the first visit	1.1	1

Appendix B: Expert Review

The external reviewers followed the steps in our brief (create an account, check the sample lease, review a High Risk clause, find what the colours mean, draft a landlord email, download the PDF) and recorded each problem against Nielsen's 10 heuristics with a severity from 0 to 4. They found 16 problems: 2 major, 5 minor and 9 issues. Most concerned consistency (H4: 9) and minimalist design (H8: 4).

Table B1. Expert review findings.

#	Where	Problem	Heuristic	Severity
1	Home page	Hero chips differ in height and width	H4	1
2	Home page	Animation should not touch the hero text	H8	1
3	Home page	Arrows on "How it works" and "Get started" differ in size	H4	1
4	Home page	Mixed Title Case and sentence case on buttons and links	H4	2
5	Home page	Too few sections to build trust (e.g. who is behind it)	H10	2
6	Create account	The form uses only about 30% of the page	H8	2
7	Upload pop-up	No way to cancel; the user is stuck until the upload finishes	H3	3
8	Loading states	Skeletons do not match the final layout	H4	2
9	Dashboard	Centre-aligned while the rest of the site is left-aligned	H4	2
10	Page headers	Uneven spacing between chip, title and subtext	H4	1
11	Summary page	Risk counts are not coloured by risk level	H6	1
12	Key lease details	Icon does not match	H4	1

13	Deposit card	Yellow suggests a warning, but it is information	H4	3
14	“Before you sign”	Underline serves no purpose	H8	1
15	Lease review cards	Card height changes with content	H4	1
16	All cards	Corner radius slightly too large	H8	1

What worked well: the skeleton loading effect; uploading and analysing the PDF; the High / Medium / Low Risk colour coding; and the Lease review, including “Ask about this clause” once tried.

First thing to change: let the upload run in the background, or at least add a Cancel button, because the user is stuck in the pop-up until it finishes.

Appendix C: Usability Test

Four participants (P1–P4), none on the team, used the built-in sample lease within the Dublin flat scenario. ✓ = completed without help; ✗ = failed or gave up.

Table C1. Task results per participant. All four answered “No” to signing the lease as it is.

Participant	Device	1 Sign up	2 Check	3 Safe to sign?	4 Deposit	5 Colours	6 Email	7 Save PDF	Answer card
P1	Tablet	✓	✓	✓	✗	✓	✗	✓	3 of 3
P2	Phone	✓	✓	✓	✗	✓	✗	✗	2 of 3
P3	Laptop	✓	✓	✓	✓	✓	✗	✓	3 of 3
P4	Laptop	✗	✓	✓	✓	✓	✓	✓	2 of 3

Table C2. Observations and comments (quotes as recorded on the note sheets).

Participant	Where they got stuck and what they said
P1 (tablet)	Uploaded the wrong lease at first. Felt “overwhelmed with the info provided”. On the email: “Is not working, was expecting an error message instead of making me wait.” Most helpful: “the application pointed out specifically what I did not understand in the lease.”
P2 (phone)	Struggled with button sizes and Lease viewer on the phone; found the report page “not clean and difficult to navigate”. Main problem: “the phone UI look of the application.” The email draft kept loading.

P3 (laptop)	“Too many general clauses shown which I felt was of no use.” Wished the email gave “a basic draft if the server failed”. Most helpful: “The ease with which I was able to login, check the lease and understand where the issue is.”
P4 (laptop)	Spent 4 minutes on the landing page deciding whether to trust the app: “Make it a little more trusting.” The email worked after a wait. Main problem: “Application was not trustworthy enough for me.”

Appendix D: Survey Questionnaire

Google Form, six sections, about 15 minutes. Participants used the live site (www.clauseguard.ie) with the sample lease; each task section showed the task first, then its questions. All questions were required except Q8b.

Table D1. Survey questions (Q17 is the form’s final, unnumbered question).

Section	Question	Answer type
1. About you (before using the app)	Age	Short answer
	Q1. Have you ever signed a residential lease in Ireland?	Yes / No
	Q2. How confident are you in understanding your rights under RTB regulations when reviewing a lease?	1–5 (not confident at all – highly confident)
	Q3. Have you ever felt pressured to sign a lease quickly without fully understanding the terms?	Yes / No / Not sure
2. Task 1: Upload the sample lease	Q4. While the document was processing, was it clear what the system was doing?	Yes, very clear / Somewhat clear / Not clear
	Q5. Did the wait time feel acceptable for a legal review?	Yes / Borderline / No
3. Task 2: Review the High Risk and Medium Risk clauses	Q6. Looking at the first flagged clause, is the plain-English explanation clear?	1–5 (not clear at all – completely clear)
	Q7. Do you understand exactly why this clause violates RTB rules?	1–5 (not at all – completely)
	Q8. Is there any legal wording on this screen that still confuses you?	No / Yes

	Q8b. If yes, which words or phrases?	Short answer (optional)
4. Task 3: Ask the chatbot “Can my landlord actually enforce this?”	Q9. Did the chatbot’s answer directly address your concern?	1–5 (not at all – completely)
	Q10. How did the chatbot’s tone feel?	Robotic / Like a lawyer / Like a helpful advisor / Other
5. Task 4: Generate a landlord email for two flagged clauses	Q11. Does the tone balance being polite with standing up for your rights?	1–5 (too weak or too aggressive – perfect balance)
	Q12. How comfortable would you be sending this exact email to a prospective landlord?	1–5 (highly uncomfortable – highly comfortable)
6. Overall feedback	Q13. How much do you trust the legal analysis provided by ClauseGuard?	1–5 (no trust – complete trust)
	Q14. Would you use this app before signing your next rental agreement?	Yes / Maybe / No
	Q15. What was the single most confusing part of using this application today?	Free text
	Q16. Is there anything you expected the app to do that it didn’t?	Free text
	Q17. Is there anything else you would like to suggest or point out about the website or its functionalities that could be improved?	Free text